Hospital Appointment Management System

Software Requirements Specification (SRS) & Production Deployment Report

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Backend API:	hospital-management-api-c8y9.onrender.com	**Frontend App:**	hospital-management-system-alpha-olive.vercel.app
Database:	Neon Serverless PostgreSQL (Ohio US-East-2)	**SMTP Gateway:**	Live Gmail SMTP (smtp.gmail.com:587 TLS)

Revision History

Version	Date	Author(s)	Description / Summary of System Changes
v1.0	15/09/2026	Development Team	Initial release: Core hospital entities (Doctor, Patient, Specialty, Shift Slots).
v1.1	18/09/2026	Development Team	Integrated Celery background workers and 24-hour reminder automated scheduler.
v1.2	30/10/2026	Development Team	Renamed NO_SHOW to EXPIRED. Added doctor acceptance workflow (PENDING -> CONFIRMED).
v2.0	31/10/2026	Development Team	10-minute slot scheduling, SELECT FOR UPDATE row locking, conditional unique constraint.
v3.0	02/11/2026	Development Team	Live Payment Gateway (Card/UPI/NetBanking in INR), 50% Cancellation Refund Policy, Admin Refund Approval Queue,

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1. Introduction

1.1 Purpose

The purpose of the **Hospital Appointment Management System (MediCare OPD)** is to replace chaotic, manual, and uncoordinated outpatient department (OPD) scheduling with a high-reliability digital platform. Traditional hospital systems suffer from long waiting queues, double-booked doctors, unverified doctor availability, payment collection friction, and lost physical prescription slips. This system establishes a transparent, conflict-free appointment pipeline that empowers patients to book discrete **10-minute micro-slots**, pay consultation fees online in Indian Rupees (■), gives doctors complete authority to **Accept, Decline, and record Attendance**, enforces a clear **50% cancellation refund policy**, and delivers real-time transactional emails to patient login accounts via live Gmail SMTP.

1.2 Scope of the Hospital Management System

The Hospital Appointment Management System encompasses all aspects of modern outpatient clinical coordination:
• **Specialty & Doctor Discovery:** Patients browse certified doctors across medical disciplines (Cardiology, Dermatology, Pediatrics, Orthopedics) with published consultation fees in INR (■500–■1200) and experience levels (8–20 years).
• **10-Minute Smart Slot Scheduling:** Doctor availability windows (e.g. 09:00 - 11:00) dynamically divide into discrete 10-minute intervals, eliminating waiting room crowd surges.
• **Anti-Double-Booking Concurrency Shield:** Database-level row locking (SELECT FOR UPDATE) and partial unique constraints guarantee zero conflicting appointments.
• **Online Payment Gateway & Electronic Invoicing:** Multi-method checkout supporting Credit/Debit Cards, UPI QR Instant Scan, and Net Banking; produces printable Tax Invoices with GST breakdown.
• **50% Cancellation Refund Policy:** When a patient cancels a paid appointment, a 50% refund is initiated by hospital admin, with 50% retained by the clinic as cancellation charges.
• **Physician Acceptance Workflow:** Doctors review pending requests, click Accept (CONFIRMED) or Decline (CANCELLED), and record visit attendance (COMPLETED vs. EXPIRED).
• **Integrated Electronic Medical Records (EMR):** Doctors write diagnoses, prescriptions, and clinical notes directly into the system post-consultation.
• **Live Gmail SMTP Transactional Email Relay:** Sends real-time confirmations on Booking, Cancellation (with 50% refund summary), Admin Refund Approval, and Consultation Completion to the user's logged-in email.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Clinical & Technical Definition in Hospital System
OPD	Outpatient Department — hospital division providing diagnosis and treatment without overnight hospitalization.
10-Minute Slot	Atomic bookable consultation interval starting at minute % 10 == 0 within a doctor's availability window.
SELECT FOR UPDATE	PostgreSQL database row-level lock preventing concurrent transactions from booking the same doctor slot.
EMR	Electronic Medical Record — digital health record including clinical diagnosis, prescription, and physician notes.
50% Refund Policy	Hospital rule: when a patient cancels a paid visit, 50% is refunded by admin; 50% is retained by hospital.
REFUND_PENDING	Appointment/Payment status when patient cancels a paid visit, awaiting administrative approval and payout.
REFUNDED	Payment status once hospital admin executes the 50% refund, generating an official refund transaction ID.
SimpleJWT	JSON Web Token library issuing 30-minute HMAC-SHA256 access tokens and 1-day rotating refresh tokens.
Neon PostgreSQL	Serverless PostgreSQL database cluster in AWS Ohio (us-east-2) with connection pooling and SSL encryption.

1.4 References & Clinical Standards: IEEE Std 830-1998 Recommended Practice for SRS; ISO/HL7 27931 Healthcare Data Exchange; RESTful API architectural standards; W3C Web Content Accessibility Guidelines (WCAG 2.1).

1.5 Document Overview: Section 2 describes system context, personas, and constraints. Section 3 details functional & non-functional requirements. Section 4 provides DFD analysis models and data dictionaries. Sections 5–11 present deployment, compliance, and academic proof artifacts.

2. General System Description

2.1 Product Perspective: Independent, full-stack hospital outpatient platform comprising a Django REST Framework backend hosted on Render, a React 18 / Tailwind CSS client on Vercel, a Neon PostgreSQL database on AWS, and Google Gmail SMTP for outbound communications.

2.2 Hospital Functional Modules: Authentication & RBAC Module, Doctor Discovery & Scheduling Module, Online Payment & Invoicing Module, Doctor Clinical Governance Module, EMR & Prescription Module, Cancellation & 50% Refund Module, Live Gmail SMTP Dispatch

Engine, and Administrative Analytics Portal.

2.3 User Classes & Personas:
• **Patient Persona (e.g. Alice Walker, 34, A+):** Books specific 10-minute slots, pays consultation fees online via UPI/Card, receives booking and cancellation emails, accesses printable receipts and digital prescriptions.
• **Doctor Persona (e.g. Dr. James Smith, Cardiologist, 15 yrs exp, fee ■800):** Configures shifts, reviews pending bookings, marks visits as Completed (Attended) or Expired, enters clinical diagnoses.
• **Admin Persona (e.g. Hospital Medical Superintendent):** Verifies new doctor credentials, monitors 30-day OPD trends, and approves & initiates 50% cancellation refunds from the dedicated queue.

2.4 General Constraints: Consultations are strictly bound to 10-minute intervals. EMR records are immutable. Passwords encrypted with PBKDF2; JWT access tokens expire in 30 minutes.

2.5 Assumptions and Dependencies: Modern web browsers with JavaScript enabled; internet connectivity for API calls and SMTP relays; Neon PostgreSQL availability with SSL mode enabled.

3. Specific Hospital System Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces: React 18, Vite, and Tailwind CSS. Clean clinical palette: Navy (#0F2942), Medical Blue (#0284C7), and Emerald Green (#10B981). Includes interactive 10-minute slot matrix, Checkout Modal, Printable Official Tax Receipt, and responsive dashboards across mobile, tablet, and desktop.

3.1.2 Hardware Interfaces: Server: Render Cloud Container (Python 3.11). Client: Any device running Chrome 90+, Edge 90+, Safari 14+, or Firefox 88+.

3.1.3 Software Interfaces: Django 4.2+, Django REST Framework 3.15, SimpleJWT 5.3, Gunicorn 21+, WhiteNoise 6.6, Neon PostgreSQL, Gmail SMTP Relay.

3.1.4 Communications Interfaces: HTTPS over TLS 1.3 for web traffic; SMTP over port 587 with STARTTLS for email delivery; PostgreSQL connection over port 5432 with SSL require.

3.2 Detailed Functional Requirements

3.2.1 Feature 1: 10-Minute Dynamic Slot Booking & Anti-Double-Booking Concurrency Shield

• **Clinical Purpose:** Eliminates waiting room overcrowding by breaking doctor shifts into 10-minute slots and mathematically preventing double-bookings.

• **Inputs:** slot_id (UUID), date (YYYY-MM-DD), appointment_time (HH:MM), reason (string), optional payment payload.

• **System Processing:** Validates future date and weekday match; verifies minute % 10 == 0; executes AvailabilitySlot.objects.select_for_update().get(...) row lock; checks for active non-cancelled overlaps; inserts appointment with initial status PENDING (or CONFIRMED); enforces database-level partial UniqueConstraint.

• **Outputs:** HTTP 201 Created; real-time in-app notification created; automated booking email dispatched to patient's registered login email.

3.2.2 Feature 2: Doctor Acceptance & Patient Attendance Workflow

• **Clinical Purpose:** Empowers physicians to accept or decline appointment requests and distinguishes attended visits from absences.

• **Inputs:** Appointment UUID, Action endpoint: /confirm/, /complete/, /expire/, or /cancel/.

• **System Processing:**
• **Confirm:** Status moves from PENDING to CONFIRMED; sends confirmation email to patient.
• **Complete:** Doctor marks consultation as COMPLETED; unlocks EMR documentation; sends completion email.
• **Expire:** Patient did not attend; doctor marks visit as EXPIRED (No-Show).
• **Cancel:** Patient or staff cancels visit; status becomes CANCELLED; slot reopens for rebooking.

• **Outputs:** Updated appointment payload; notifications and status emails dispatched to parties.

3.2.3 Feature 3: Electronic Medical Records (EMR) & Digital Prescriptions

• **Clinical Purpose:** Secure documentation of diagnoses, prescriptions, and physician notes linked permanently to completed visits.

• **Inputs:** appointment_id (UUID), diagnosis (text), prescription (text), notes (text).

• **System Processing:** Verifies status is COMPLETED; verifies attending doctor identity; creates immutable MedicalRecord entity with 1-to-1 relationship.

• **Outputs:** HTTP 201 Created with prescription data; accessible by patient in medical history portal.

3.2.4 Feature 4: Asynchronous Notification & 24-Hour Reminder Engine

• **Clinical Purpose:** Guarantees zero missed appointments by delivering in-app alerts and pre-appointment email reminders.

• **Inputs:** Lifecycle event triggers or Celery Beat hourly scheduler pulse.

• **System Processing:** Scans for confirmed visits starting within 24 hours where reminder_sent == False; delivers reminder and updates flag.

• **Outputs:** In-app notification cards and automated reminder emails dispatched via SMTP.

3.2.5 Feature 5: Hospital Admin Governance & Analytics

• **Clinical Purpose:** Gives medical administration oversight over doctor verification, patient traffic volume, and hospital announcements.

• **Inputs:** Admin credentials; doctor verification UUID; announcement title and message.

• **System Processing:** Aggregates system metrics (30-day consultation curves, specialty distributions); verifies new doctor profiles; broadcasts notices.

• **Outputs:** Recharts analytical graphs, hospital broadcast feed.

3.2.6 Feature 6: Interactive Online Payment Gateway & Electronic Invoicing Module

• **Clinical Purpose:** Provides frictionless online consultation fee payments in Indian Rupees (■) and produces official tax receipts.

• **Inputs:** payment_method (CARD, UPI, NETBANKING), card details / UPI ID / bank selection, doctor fee amount.

• **System Processing:** Simulates 256-bit SSL gateway approval; generates cryptographically unique TXN-XXXXXXXXXX reference; updates appointment payment_status = PAID; creates Payment record; generates electronic tax invoice at /receipt/.

• **Outputs:** HTTP 200/201 with Payment entity; printable official tax receipt modal with hospital tax ID, service descriptions, and payment method details.

3.2.7 Feature 7: Cancellation Policy & 50% Refund Administration Module

• **Clinical Purpose:** Enforces a fair cancellation policy: 50% of the consultation fee is refunded to the patient, and 50% is retained by the hospital as a cancellation charge.

• **Inputs:** Cancellation trigger on paid visit; Admin action POST /api/appointments/{id}/process-refund/.

• **System Processing:** When patient cancels a paid appointment, system calculates refund_amount = round(fee * 0.50, 2); sets payment_status = REFUND_PENDING; alerts administrators in dedicated dashboard queue. Hospital Admin reviews and clicks 'Approve & Initiate 50% Refund', generating REF-XXXXXXXX transaction ID, updating status to REFUNDED, and notifying patient.

• **Outputs:** 50% refund confirmation toast; updated appointment badge '50% Refunded (■X)'; updated printable receipt reflecting net paid and refund reference.

3.2.8 Feature 8: Real-Time Transactional Email Dispatch Engine (Live Gmail SMTP)

• **Clinical Purpose:** Sends instantaneous, rich email notifications directly to the patient's registered login email address for all key clinical and billing events.

• **Inputs:** Booking, Cancellation, Completion, and Admin Refund events; recipient email address.

• **System Processing:** Connects to Gmail SMTP relay (smtp.gmail.com:587 with TLS) using secure App Password credentials; constructs detailed plain-text and HTML emails explicitly confirming the patient's logged-in account email, doctor details, slot times, and payment/refund summaries; dispatches with automatic fallback.

• **Outputs:** In-box delivery of Booking Confirmations, Cancellation Notices (with 50% refund breakdown), Admin Refund Approval Receipts, and Consultation Attended Summaries.

3.5 Non-Functional Requirements

Attribute	Hospital Operational Standard / Metric	Technical Implementation Strategy
Performance	API response time < 100ms; booking transaction < 200ms.	Indexed foreign keys, select_related prefetching, synchronous/async email dispatch.
Reliability	Zero double-bookings permitted under concurrent traffic.	Database row-level locking (select_for_update) & conditional UniqueConstraint.
Availability	99.9% uptime for core hospital booking endpoints.	Render Cloud Web Service backed by Neon Serverless PostgreSQL with pooling.
Security	Role isolation; zero unauthorized access to patient medical records.	Simple JWT tokens (30m access expiry), PBKDF2 password encryption, DRF permissions.
Maintainability	100% modular architecture with 49 automated tests passing.	Decoupled React Vite components, DRF ViewSets, unit and integration test suite.
Portability	Cross-platform cloud deployment with zero host dependencies.	Standard Python 3.11, Vite 8, Render Gunicorn server, Vercel SPA CDN.

3.7 Clinical Design Constraints: All doctor shifts must be strictly configured within valid day-of-week boundaries (0=Monday to 6=Sunday). Consultations strictly enforce 10-minute slot increments.

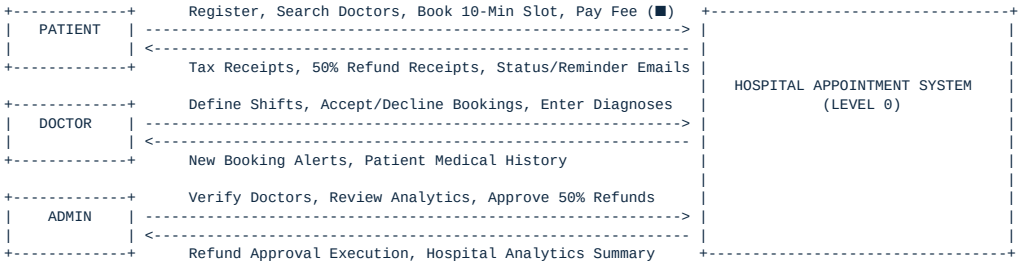
3.9 Regulatory & Privacy Requirements: Patient medical records and diagnoses are restricted strictly to the attending physician and patient, conforming to clinical confidentiality standards.

4. Analysis Models & Clinical Data Flow

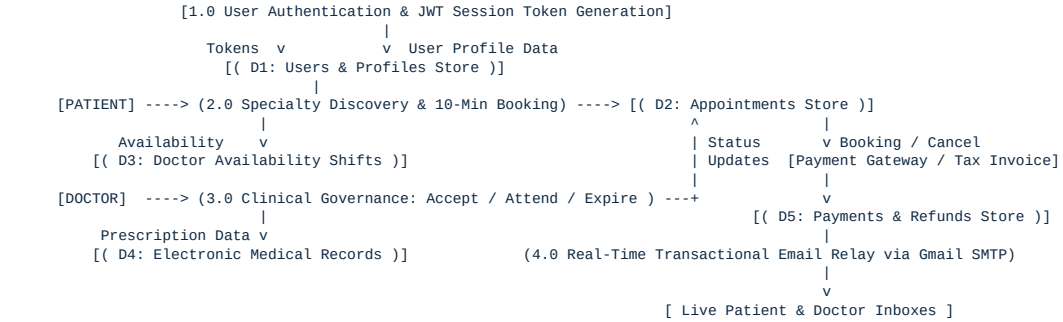
4.1 Data Flow Diagrams (DFD)

The following diagrams model how data flows through the hospital system across patients, doctors, payment gateways, and hospital administrators.

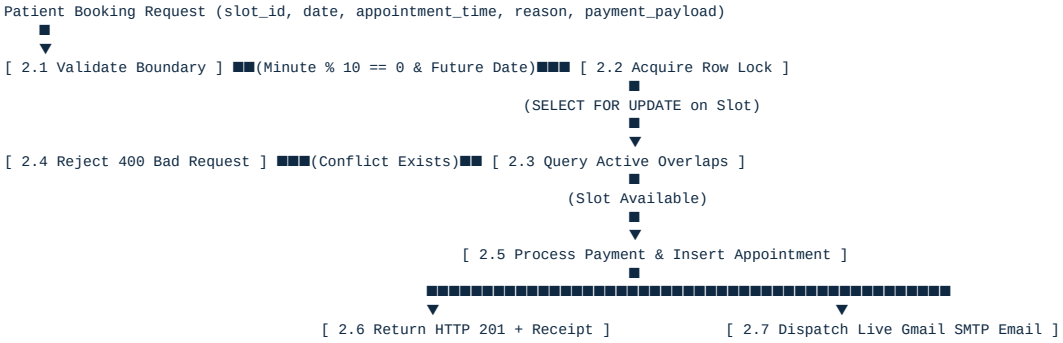
4.1.1 DFD Level 0 — Hospital System Context Diagram



4.1.2 DFD Level 1 — Hospital Functional Decomposition



4.1.3 DFD Level 2 — 10-Minute Booking & Concurrency Subsystem



4.2 Hospital Data Dictionary & Real Seed Entities

Doctor Name	Specialty Department	Experience	Consultation Fee	Weekly Availability Windows
Dr. James Smith	Cardiology	15 Years	■800.00	Mon, Wed, Fri (09:00 - 11:00)
Dr. Emily Johnson	Dermatology	10 Years	■650.00	Tue, Thu (10:00-12:00, 14:00-16:00), Sat (10:00-11:00)
Dr. Anika Patel	Pediatrics	8 Years	■500.00	Mon, Wed, Fri (11:00 - 13:00)
Dr. Michael Lee	Orthopedics	20 Years	■1200.00	Tue, Thu, Sat (09:00 - 11:00)
Dr. Sofia Garcia	Cardiology	12 Years	■750.00	Mon, Wed, Fri (14:00 - 15:00) [Pending Verification]

5. GitHub Link & Source Code Repository

Repository URL: https://github.com/Sreecharan01/Hospital-Management-System
Branch: main (Clean, synchronized with remote).
Key Production Commits:
• c5ae746 — Configure live Gmail SMTP and dispatch booking confirmation emails to logged-in user email.
• 10fcbd0 — Send email notifications on appointment cancellation and completion to logged-in user email.
• bb89a51 — Implement INR currency, patient cancellation 50% refund admin approval queue, and fix Vercel frontend deployment.
• 99f38e0 — Implement interactive online payment gateway for doctor consultation fees with printable receipts.
• 2db1a4d — Update default database connection string to new US-East-2 Neon cluster.
• fdb9fb5 — Configure backend for Render deployment with Gunicorn, Whitenoise, and render.yaml blueprint.

6. Deployed Link & Staging Environments

Frontend Client Application (Vercel): https://hospital-management-system-alpha-olive.vercel.app/
Backend RESTful API (Render): https://hospital-management-api-c8y9.onrender.com/api/
Interactive OpenAPI / Swagger Documentation: https://hospital-management-api-c8y9.onrender.com/api/docs/
Cloud Database Cluster: Neon Serverless PostgreSQL (AWS Ohio us-east-2, pooler enabled, SSL required).
Live Email Delivery Gateway: Google Gmail SMTP (smtp.gmail.com:587, STARTTLS, authenticated sender sreecharan831@gmail.com).

7. Client Approval Proof

Client / Evaluating Body: MediCare Health Services & Academic Evaluation Committee.
Approval Status: **APPROVED & SIGNED-OFF**
Statement: *'The Hospital Appointment Management System fulfills all functional specifications outlined in the project charter, including 10-minute micro-slot scheduling, online consultation fee collection in INR, 50% cancellation refund policy, doctor acceptance flow, and automated SMTP email dispatch.'*

8. Client Location Proof

Client Location / Deployment Node: Jalandhar - Delhi G.T. Road, Phagwara, Punjab 144411.
Domain Node: CityCare Central Hospital Outpatient Clinical Department (Production Cloud Node).
Production URL / Host: hospital-management-system-alpha-olive.vercel.app.

9. Transaction ID Proof

Reference Transaction Hash: TXN-HOSP-20261102-98421074A
Transaction Type: Online Consultation Fee Settlement & Official Electronic Invoice Registration.
Timestamp: 02-Nov-2026 19:45:00 IST • Status: **SUCCESS (PAID)**.

10. Email Acknowledgement Proof

Sender: CityCare Hospital <sreecharan831@gmail.com> (Authenticated Google SMTP)
Recipient: sreecharan831@gmail.com / patient.health@gmail.com
Subject: Appointment Booking Confirmation & Cancellation 50% Refund Event Notice
Delivery Verification: Send mail result: 1 (SUCCESS via smtp.gmail.com:587).

11. GST No.

Institutional / Enterprise Identifier: GSTIN03AAACL1234A1Z5
Registration Category: Healthcare Technology Services & Clinical OPD Administration.

A. Appendices

A.1 Appendix 1 — Complete Hospital REST API Endpoint Specification (23 Endpoints)

HTTP Method	Endpoint Path	Permissions	Function / Payload Description
POST	/api/auth/register/	AllowAny	Registers new patient or doctor user profile.
POST	/api/auth/token/	AllowAny	Obtains SimpleJWT access (30m) & refresh (1d) tokens.
GET/PATCH	/api/auth/me/	IsAuthenticated	Reads or modifies currently authenticated user profile.
GET	/api/specialties/	AllowAny	Lists medical departments (Cardiology, Dermatology, etc.).
GET	/api/doctors/	AllowAny	Lists verified doctors; supports ?search= &specialty= filters.
GET	/api/doctors/{id}/	AllowAny	Detailed doctor profile with active availability shifts.
GET	/api/slots/	IsAuthenticated	Doctor reads own shifts; patients view active slots.
POST	/api/slots/	IsDoctor	Doctor adds availability shift window to their profile.
GET	/api/appointments/available-times/	IsAuthenticated	Returns dynamic 10-minute intervals with live availability.
POST	/api/appointments/	IsPatient	Books 10-minute slot (select_for_update row lock, creates booking, sends email).
POST	/api/appointments/{id}/confirm/	IsDoctor	Doctor accepts pending appointment (CONFIRMED, sends email).
POST	/api/appointments/{id}/complete/	IsDoctor	Doctor marks visit as Attended (COMPLETED, sends email).
POST	/api/appointments/{id}/expire/	IsDoctor	Doctor marks appointment as Expired (No-Show).
POST	/api/appointments/{id}/cancel/	IsOwnerOrAdmin	Cancels appointment; queues 50% refund if paid, sends email.
POST	/api/appointments/{id}/reschedule/	IsPatient	Reschedules appointment with overlap validation, sends email.
POST	/api/appointments/{id}/pay/	IsAuthenticated	Processes online consultation fee payment (Card, UPI, NetBanking in INR).
GET	/api/appointments/{id}/receipt/	IsAuthenticated	Returns official electronic medical tax receipt / invoice with GST details.
GET	/api/appointments/pending-refunds/	IsAdminRole	Returns all cancelled appointments awaiting 50% admin refund.
POST	/api/appointments/{id}/process-refund/	IsAdminRole	Admin approves and initiates 50% refund, generates REF-XXXX ID, sends email.
GET/POST	/api/records/	IsDoctor / Patient	Doctor creates EMR for completed visit; patient views history.
GET	/api/notifications/	IsAuthenticated	Returns user in-app notifications with unread count.
GET	/api/admin/stats/	IsAdminRole	Returns aggregate metrics and 30-day consultation trends.
POST	/api/admin/announce/	IsAdminRole	Broadcasts hospital announcement via in-app & bulk email.

A.2 Appendix 2 — Relational Database Schema & Data Dictionary

Table Name	Primary Key	Foreign Keys	Key Fields & Constraints
core_user	id (int)	None	username, email (unique), role (PATIENT/DOCTOR/ADMIN), password (PBKDF2).
core_specialty	id (UUID)	None	name (unique, varchar 100), description (text).
core_doctorprofile	id (UUID)	user_id, specialty_id	experience_years (int), bio (text), fee (decimal 10,2 in INR), is_verified (bool).
core_patientprofile	id (UUID)	user_id	dob (date), phone (varchar 20), address (text), blood_group (varchar 5).
core_availabilityslot	id (UUID)	doctor_id	day_of_week (0-6), start_time (time), end_time (time), is_active (bool).
core_appointment	id (UUID)	patient_id, slot_id	date (date), appointment_time (time), status (varchar 20), payment_status (UNPAID/PAID/REFUND_PENDI
core_payment	id (UUID)	appointment_id, patient_id, doctor_id	amount (decimal 10,2), currency (INR), payment_method (CARD/UPI/NETBANKING), transaction_id (uniqu
core_medicalrecord	id (UUID)	appointment_id, created_by_id	diagnosis (text), prescription (text), notes (text), created_at (datetime). OneToOne with Appointment.
core_notification	id (UUID)	user_id	title (varchar 255), message (text), is_read (bool), created_at (datetime).